



Passive Solar Tracking System

ME 325 – Third Year Engineering Project

Group Members - E/21/114
E/21/387
E/21/446

Group number - 08

Supervisor - Ms.L.U.Bakmeedeniya

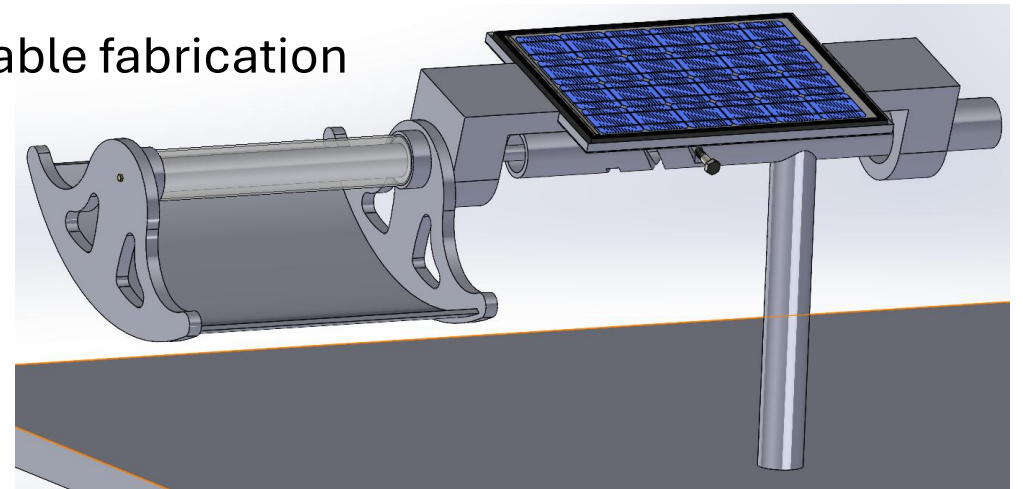


Main Objective

- Design and develop a fully passive single-axis solar tracker using paraffin wax expansion.

Specific Objectives

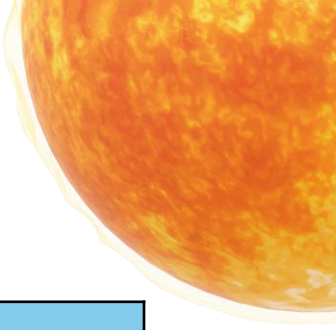
- Track sun for 8 hours
- No electricity consumption
- Maximum tracking lag $< 8^\circ$
- Low maintenance
- Suitable for Sri Lankan climate
- Affordable fabrication



Project Timeline

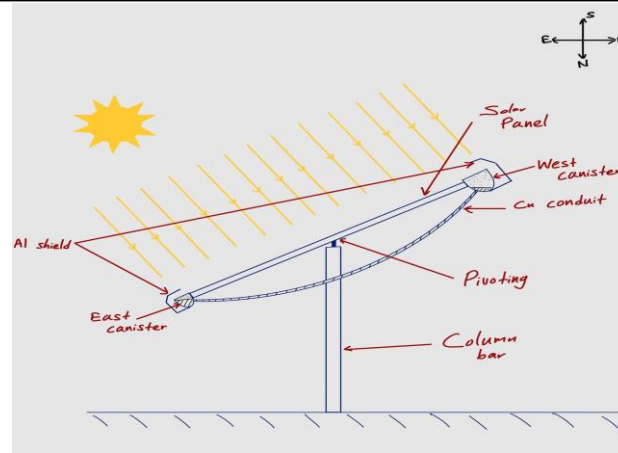
Milestone	Due	Completion Status	Remarks
Literature Review & Patent Analysis	Week 2	100%	Completed review of research papers about Passive Solar Trackers
Concept Selection & Design Methodology	Week 4	100%	Paraffin wax system selected over refrigerant based system
Mathematical Modeling	Week 6	100%	Thermal, phase change and mechanical models developed
CAD Design & System Architecture	Week 8	90%	Main components modeled; minor refinements ongoing
Excel/MATLAB Simulation	Week 10	85%	Tracking angle and lag analysis completed
Prototype Fabrication Planning	Week 12	70%	Material selection and fabrication drawings prepared



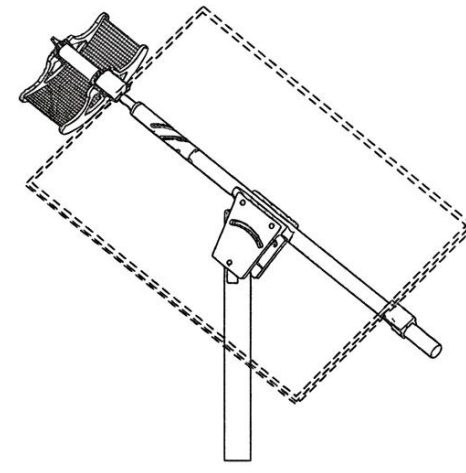
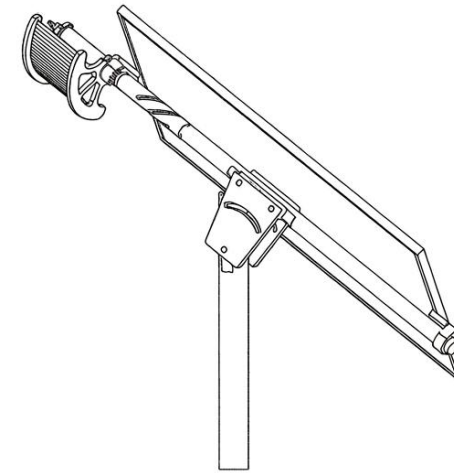
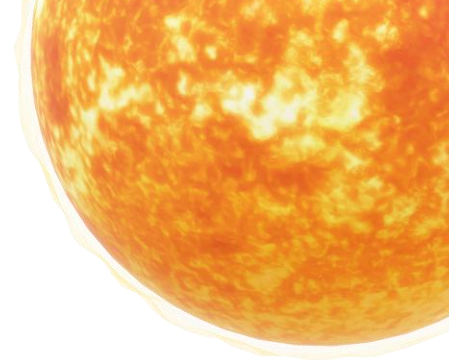
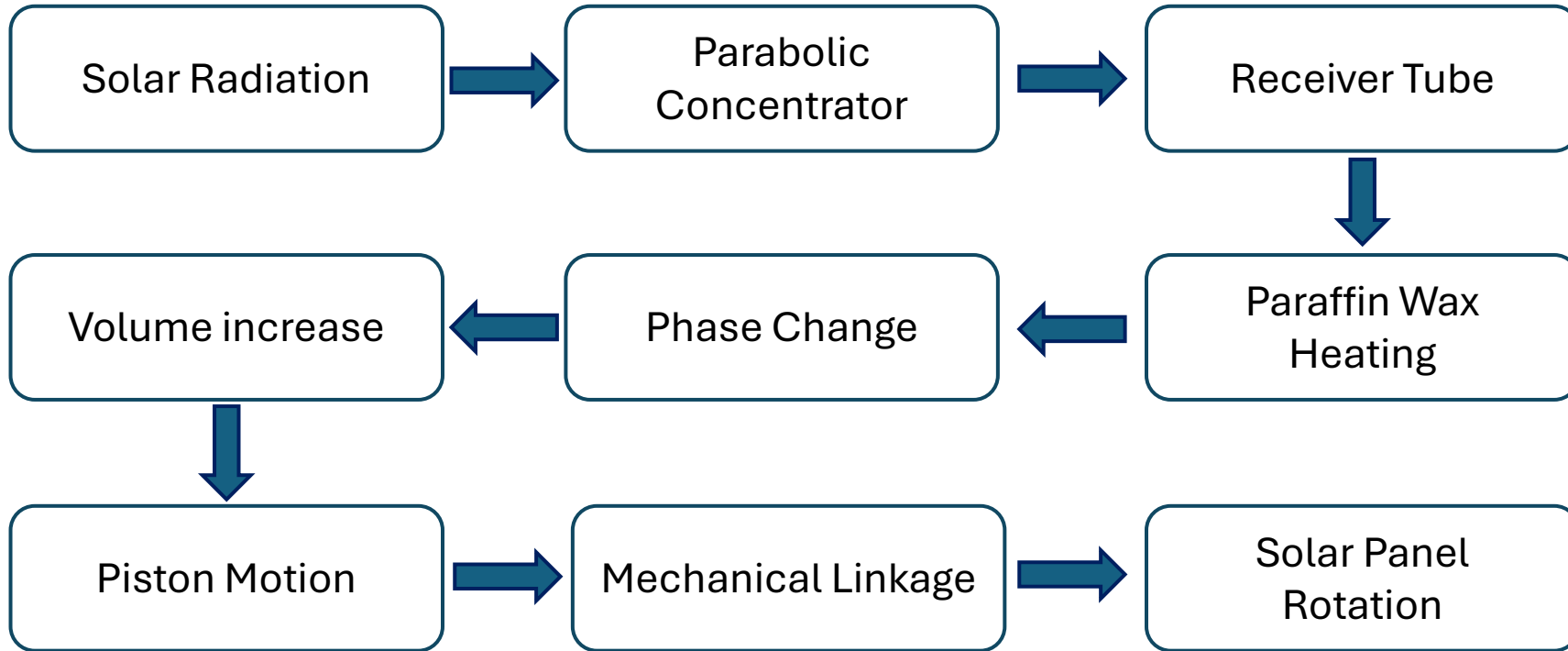


Concept Comparison

Criterion	Refrigerant System	Paraffin Wax System
Safety	High pressure fluid	Low pressure operation
Leakage Risk	High	Very low
Manufacturing Complexity	High	Simple
Sealing Requirements	Critical	Moderate
Maintenance	High	Low
Cost	Higher	Lower
Availability	Refrigerant procurement required	Easily available

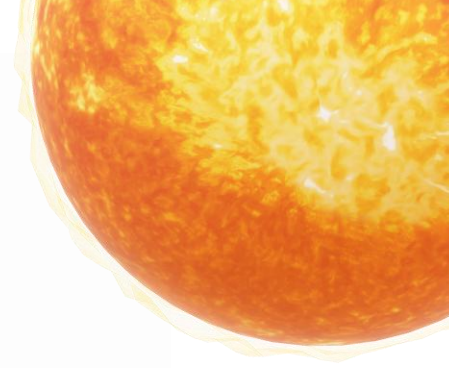


Design Methodology



Given Data

- Aperture Width = 0.50 m
- Wax Length = 0.2387 m
- Wax Volume = 300 mL = 0.0003 m^3
- Piston Diameter = 4 cm = 0.04 m
- Piston Area = $1.257 \times 10^{-3} \text{ m}^2$
- Optical Efficiency = 0.85
- Wax Density = 900 kg/m^3
- Specific Heat = 2500 J/kgK
- Melting Temperature = 58° C
- Initial Temperature = 30° C
- Latent Heat = 200000 J/kg
- Effective Wax Volume Change = 15%



Aperture area

$$A = 0.5 \times 0.2387 \\ = 0.11935 \text{ m}^2$$

Wax mass

$$m = 900 \times 0.0003 \\ = 0.27 \text{ kg}$$

Energy to reach Melting

$$Q = m C_p \Delta T \\ = 0.27 \times 2500 \times (58 - 30) \\ = 18900 \text{ J}$$

Effective volume change.

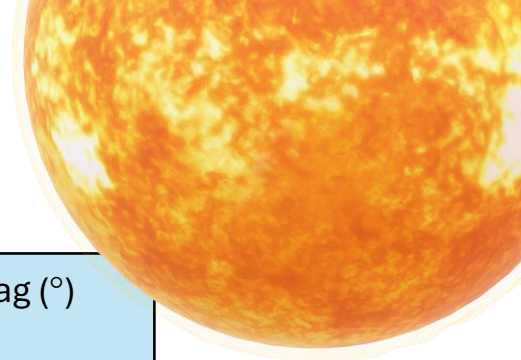
$$\Delta V = 0.15 \times 0.0003 \\ = 4.5 \times 10^{-5} \text{ m}^3$$

Maximum piston stroke.

$$x = \Delta V / A_p \\ = \frac{4.5 \times 10^{-5}}{1.257 \times 10^{-3}} \\ = 35.8 \text{ mm}$$

Required Mechanical Amplification

$$\frac{100^\circ}{35.8 \text{ mm}} = 2.79^\circ / \text{mm}$$



Results

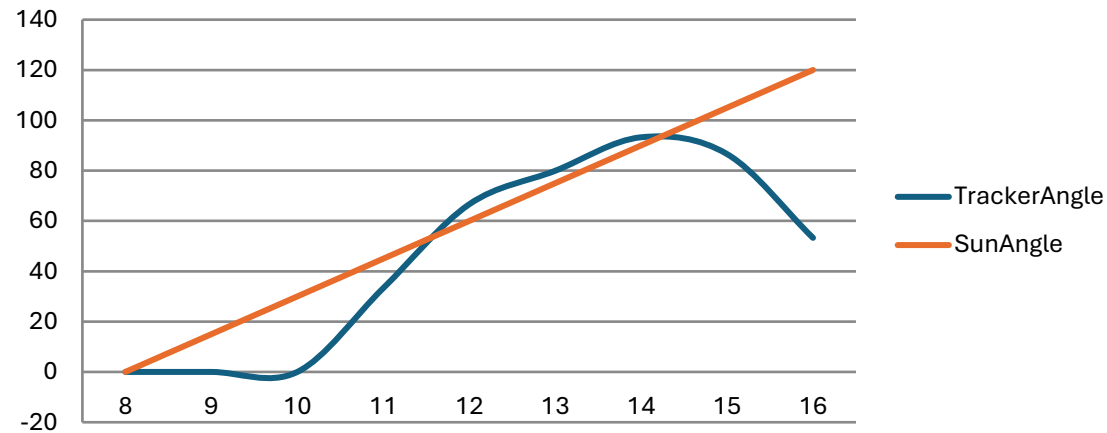
Time	Solar Irradiance	Heat Input (W)	Heat Loss (W)	Wax Temp (°C)	Melt Fraction	Stroke (mm)	Tracker Angle (°)	Lag (°)
8 AM	400	40.6	0.0	30	0.0	0.0	0	0
9 AM	600	60.9	0.1	45	0.1	3.58	10	5
10 AM	750	76.1	0.2	55	0.25	8.95	25	5
11 AM	850	86.2	0.3	58	0.4	14.32	40	5
12 PM	900	91.3	0.38	60	0.6	21.48	60	0
1 PM	850	86.2	0.35	62	0.75	26.85	75	0
2 PM	700	71.0	0.25	64	0.9	32.22	90	0
3 PM	500	50.7	0.15	65	1.0	35.8	100	5



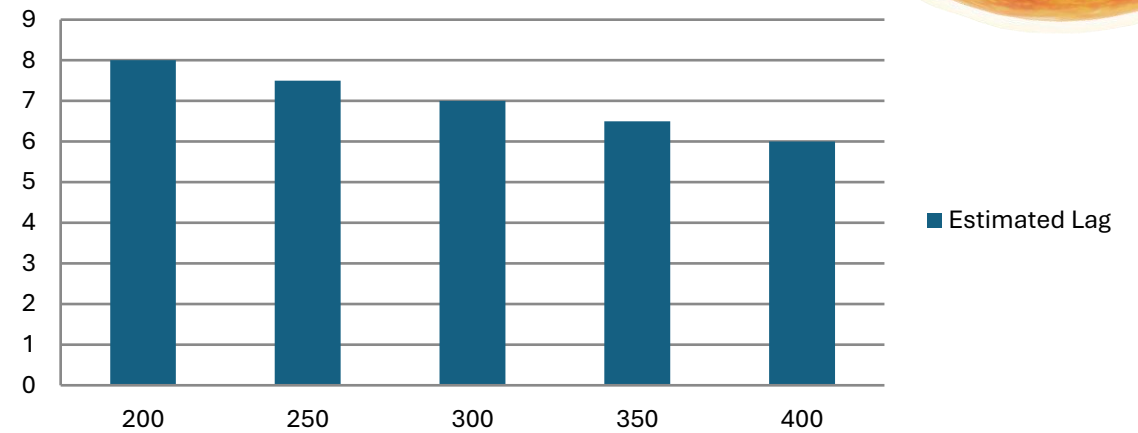
- ✓ Wax temperature increased from **30°C to 65°C**
- ✓ Full wax melting achieved by **3 PM**
- ✓ Tracker rotated from **0° to 100°**

Preliminary Results

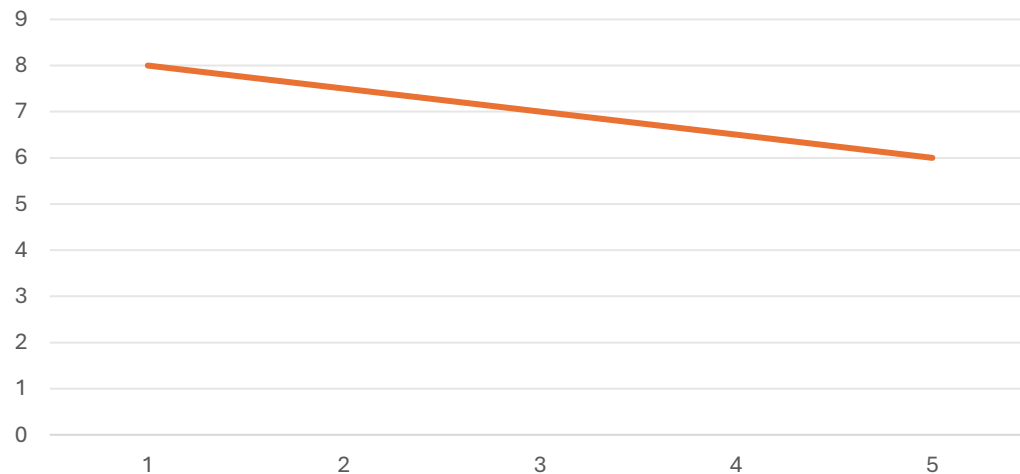
Sun vs Tracker Angle



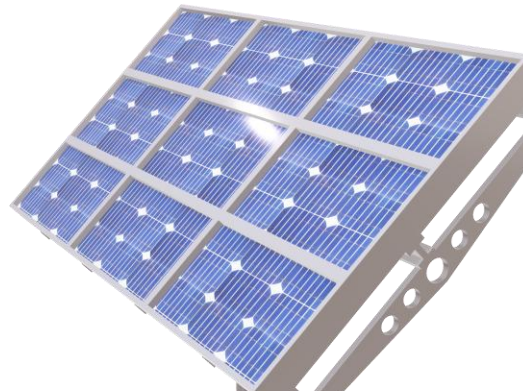
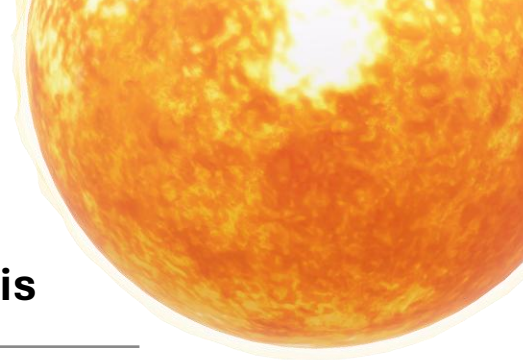
Sensitivity Analysis

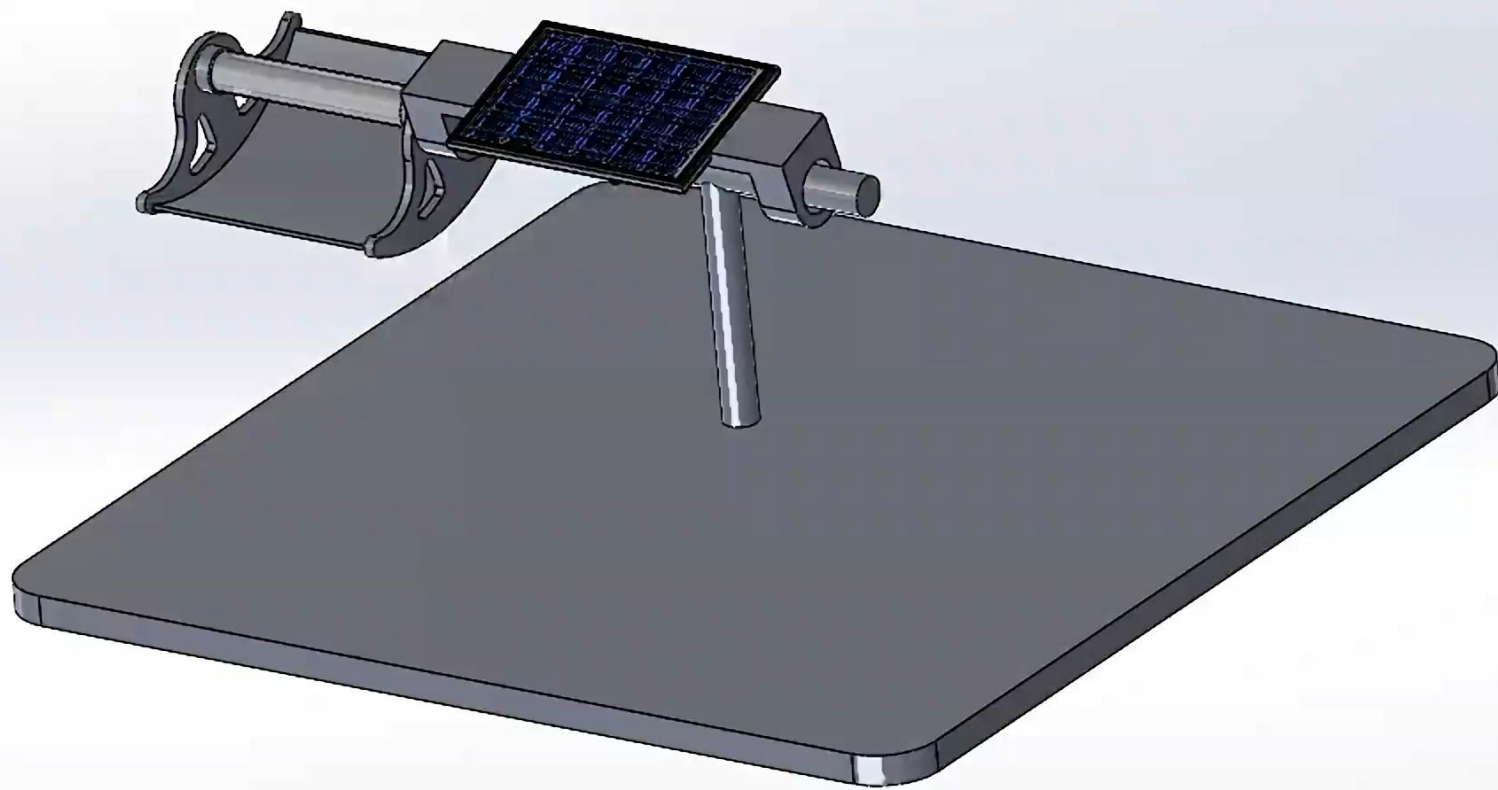


Wax Volume Vs Estimated Lag



- Maximum Lag = 7°
- Tracking Range $\approx 120^\circ$
- Passive tracking successfully achieved
- Design validated for prototype development







Thank You

Q & A

